
Detector-Plane Imaging: A Measurement-Plane Framework for Coherence Dynamics

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Abstract

We introduce *Detector-Plane Imaging* (DPI), a measurement-plane framework that treats the detector surface as an active imaging geometry capable of recording evolving spatial coherence prior to detector discretization. Using simulations of nonlinear scalar fields that support long-lived coherent excitations, we demonstrate how structured imprint fields form, evolve, and decohere, leaving characteristic spatial signatures directly on the detector plane. These imprint fields generate high-visibility interference when used as illumination in classical double-slit configurations and exhibit predictable loss of contrast under controlled decoherence.

All results are obtained within a purely classical field framework. DPI does not model photon quantization or quantum state reduction. Instead, it provides a coherence-centric interpretation of detector-plane measurements, showing that many morphological and statistical features commonly associated with single-photon interference experiments arise naturally from pre-discretization field coherence. DPI reframes measurement as a continuous dynamical process and establishes the detector plane as a direct diagnostic of coherence evolution.

1. Introduction

Time-resolved interference experiments are commonly interpreted as the accumulation of discrete detection events, where spatial structure emerges only after many localized detections. However, photon-sparse experiments demonstrate that initially noisy detection patterns gradually converge toward high-visibility interference through cumulative buildup (Aspden et al., 2016; Kolenderski & Wasilewski, 2009).

Detector-Plane Imaging (DPI) proposes a complementary interpretation. DPI treats the detector plane as a *measurement geometry* that directly samples spatial coherence prior to detector discretization. From this perspective, fringe visibility, spatial modulation, and their decay are encoded in the evolving structure of the field at the detector plane itself, rather than arising solely from post-hoc statistical aggregation.

This work demonstrates DPI using classical nonlinear field simulations. The goal is not to replace quantum theory, but to show that coherence dynamics alone are sufficient to generate detector-plane structures closely resembling those observed in interference experiments traditionally described in single-photon terms.

2. Conceptual Scope and Limitations

This manuscript operates strictly within a classical field regime. DPI does not address photon quantization, entanglement, nonlocal correlations, Bell-inequality violations, or quantum state reduction.

The term *pre-measurement* is used operationally to denote field evolution prior to detector discretization and does not assert hidden ontological states. DPI is best understood as a coherence-imaging framework consistent with classical and statistical optics (Mandel & Wolf, 1995; Goodman, 2015).

3. Simulation Framework

3.1 Field Dynamics

We simulate a nonlinear scalar field $\phi(x, y, z, t)$ governed by a ϕ^4 -type potential supporting localized coherent excitations. The field is discretized on a spatial lattice and evolved using standard numerical schemes. Boundary conditions suppress artificial reflections.

3.2 Detector-Plane Projection

A detector plane is defined at fixed $z = z_0$. At each time step, the detector records the instantaneous intensity

$$I(x, y, t) = \phi^2(x, y, z_0, t).$$

No imaging optics are applied. The detector plane acts as a passive sampling surface analogous to camera sensors used in optical experiments.

3.3 Accumulated Intensity

Long-exposure acquisition is modeled as

$$\bar{I}(x, y) = (1/N) \sum I(x, y, t_n),$$

reproducing the gradual emergence of structured patterns observed in interference buildup experiments (Aspden et al., 2016).

4. Coherence Diagnostics

DPI employs standard coherence measures evaluated explicitly at the detector plane:

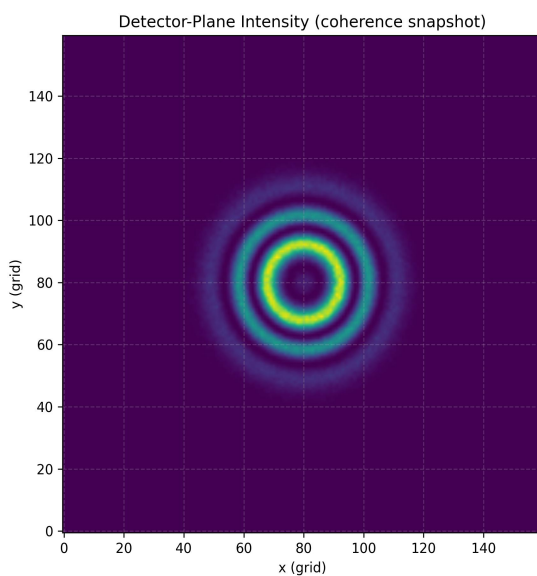
- Two-point correlation function $C(r)$
- Correlation length $\xi(t)$ extracted from $C(r)$
- Fourier structure factor $S(k)$
- Envelope width $\sigma(t)$ defined via the RMS radius of the intensity distribution

These diagnostics are well established in coherence and statistical optics (Mandel & Wolf, 1995; Goodman, 2015). DPI's contribution lies in their measurement-plane interpretation.

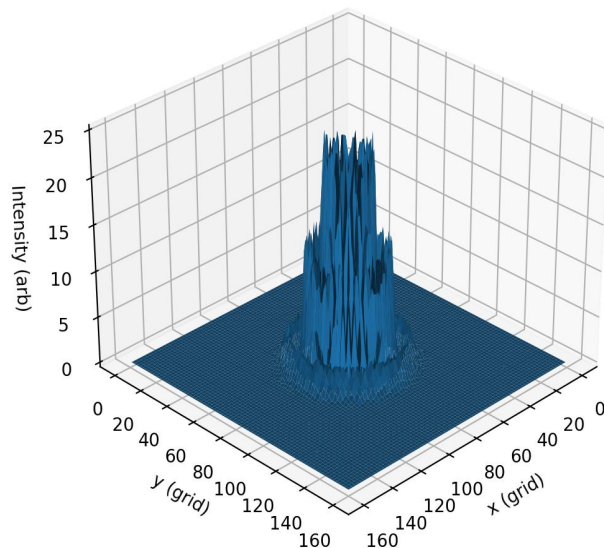
5. Coherence and Decoherence Dynamics

When phase synchronization is sustained, the detector plane exhibits smooth concentric coherence rings, long-range spatial correlations, narrow spectral peaks, and stable localized oscillatory structures. These features indicate extended coherence and low modal scattering.

As nonlinear interactions redistribute energy across modes, correlation length contracts, Fourier spectra broaden, coherence rings fragment, and detector-plane textures become stochastic. Decoherence appears as a continuous dynamical process rather than an abrupt transition, consistent with classical coherence theory (Goodman, 1976).



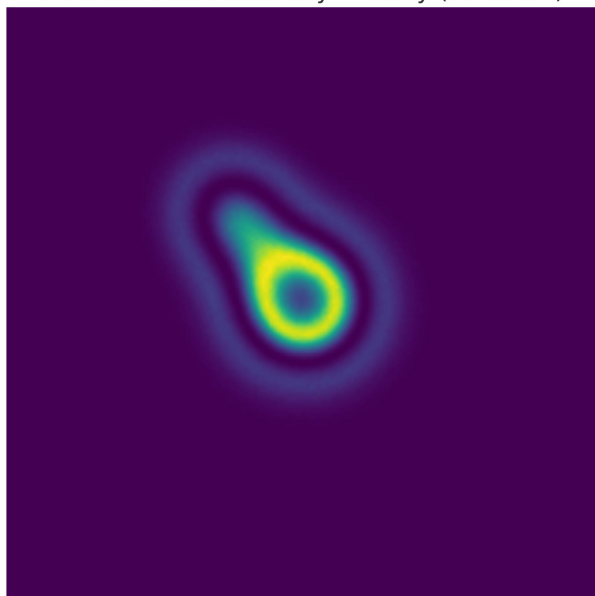
Imprint Field — pseudo-3D (frame=45)



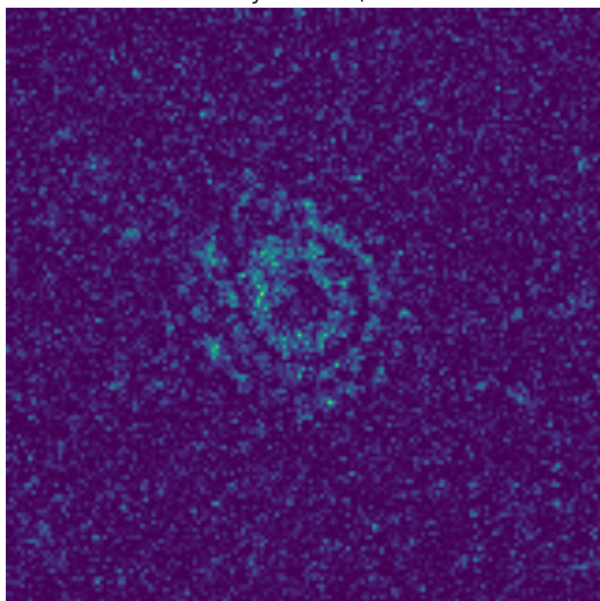
5.1 Coherence Dynamics

5.2 Decoherence Dynamics

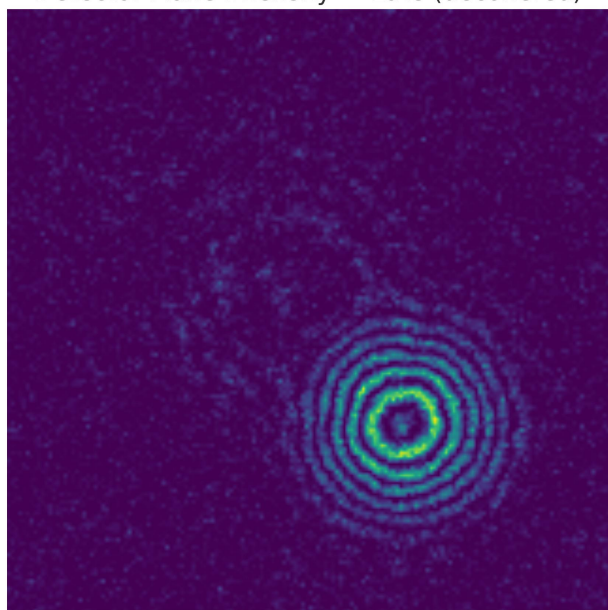
Detector-Plane Intensity — Early (coherent)



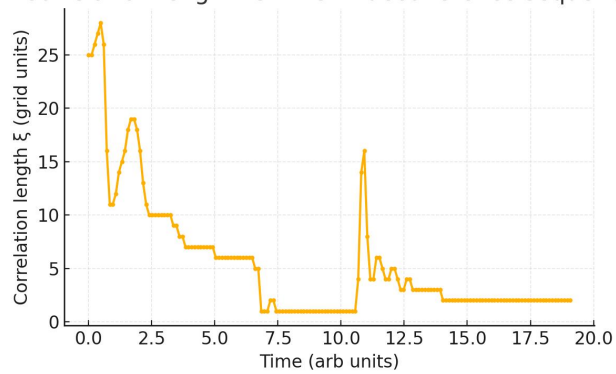
Detector-Plane Intensity — Mid (onset of decoherence)



Detector-Plane Intensity — Late (decohered)



Correlation length vs time — decoherence sequence

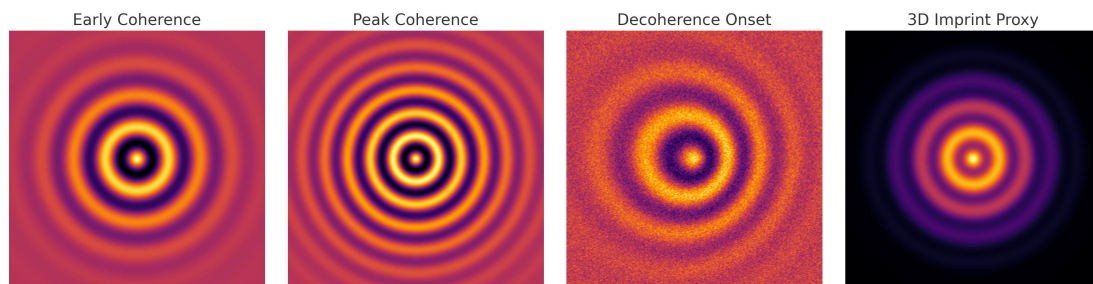


6. Double-Slit Interference Using DPI Fields

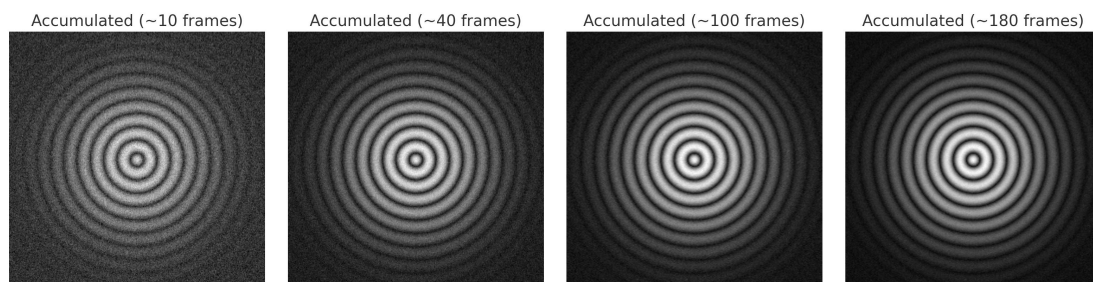
To test whether detector-plane imprint fields carry physically meaningful coherence, snapshots are used as illumination sources for a classical double-slit aperture.

When coherence spans the slit separation, far-field propagation yields high-visibility interference fringes with deep minima and structured diffraction envelopes. Introducing controlled phase noise reduces mutual coherence, lifting minima and suppressing fringe contrast in agreement with standard visibility theory (Mandel & Wolf, 1995).

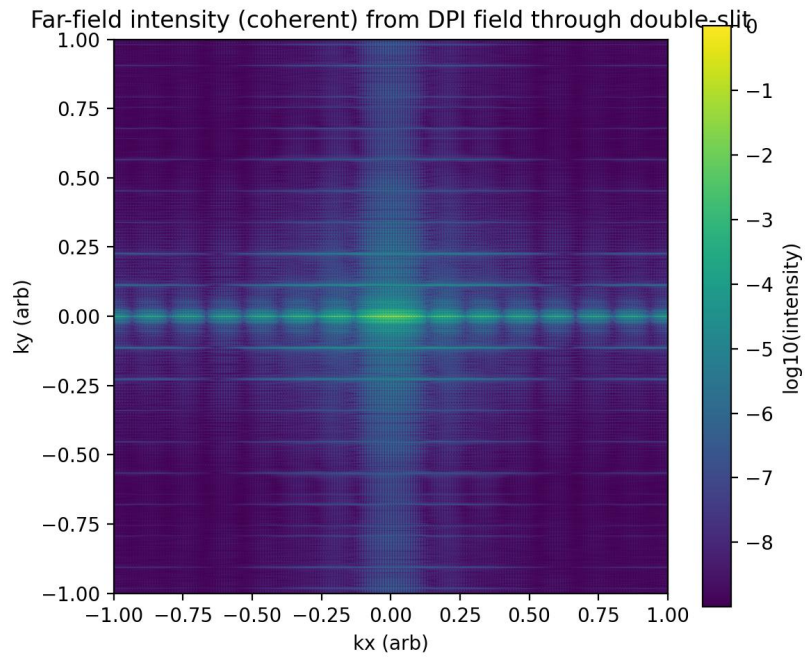
6.1 Detector-plane imprint evolution



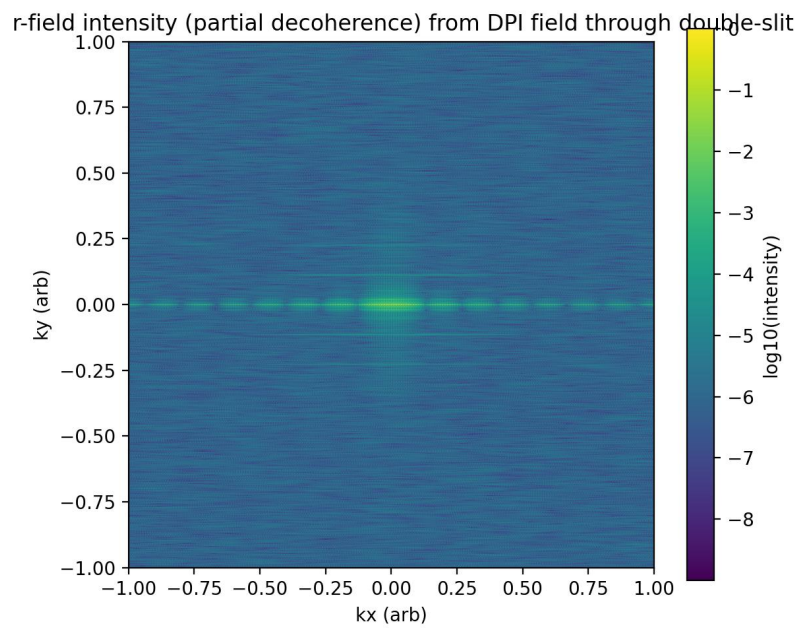
6.2 Accumulated intensity buildup



6.3 Double-slit interference (coherent)



6.4 Double-slit interference (decoherent)



7. Oscillon Analogy and Its Limits

Localized oscillatory structures in nonlinear fields provide a geometric analogy for coherent wave packets: spatial localization, phase coherence, and envelope–carrier dynamics. This analogy is strictly non-ontological. Oscillons are not photons, and DPI does not propose a sub-quantum model. The analogy clarifies why structured coherence naturally imprints on detector planes without invoking particle-based assumptions.

8. Experimental Realizability

DPI requires no specialized instrumentation. Any sensor measuring intensity at a fixed plane implements DPI geometry, including ICCD, EMCCD, SPAD arrays, sCMOS sensors, and consumer CMOS cameras.

A falsifiable prediction of DPI is that time-gated detector-plane correlation length $\xi(t)$ depends on exposure window even when accumulated intensity is fixed. Intensity-only interpretations predict no such dependence.

9. Interpretation

DPI demonstrates that detector planes encode evolving coherence structure rather than merely recording discrete detection events. Interference patterns reflect real spatial order present at the measurement plane prior to discretization, accessible through measurement-plane diagnostics.

10. Conclusion

Detector-Plane Imaging establishes the detector plane as a coherence-sensitive imaging surface. Using classical nonlinear field dynamics, DPI reproduces key spatial and temporal features associated with interference experiments, including fringe formation, visibility decay, and cumulative buildup.

By shifting emphasis from detection events to measurement-plane coherence, DPI provides a unified framework applicable across classical and quantum-optical platforms and motivates future work on coherence-aware imaging and detector design.

References

Aspden, R. S., Padgett, M. J., & Spalding, G. C. (2016). *Photon-sparse microscopy: visible light imaging using infrared illumination*. Scientific Reports, 6, 38695.

Kolenderski, P., & Wasilewski, W. (2009). *Coherence properties of heralded single photons from spontaneous parametric down-conversion*. Physical Review A, 80, 013811.

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